WENJIA GE

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SUMMARY

Electrical engineer with a physics background in sensors, embedded hardware testing, and signal analysis. Built multi-sensor data acquisition, ran 24/48-hour stability tests, and validated 7 robotic subsystems.

EDUCATION

New York University, Tandon School of Engineering New York, NY
M.S. in Electrical Engineering, GPA 3.93/4.00; Mentor Scholarship *Sep 2024 – Dec 2026*
Teaching Assistant, Digital Signal Processing I (Fall 2026). Coursework: DSP, Real-Time Embedded Systems, Machine Learning, Probability and Stochastic Processes, Electromagnetic Theory, Physics of Nanoelectronics

Shenzhen University Shenzhen, China
B.S. in Physics, GPA 3.65/4.00; Outstanding Undergraduate Thesis Award *Sep 2020 – Jul 2024*
Coursework: Optics, Information Photoelectronics, Thin Film Physics and Technology, Solid-State Physics, Analog Circuits

EXPERIENCE

AI^2 Robotics Shenzhen, China
Research Intern, Embodied AI and Robotics *Oct 2025 – Jan 2026*

- Designed test protocols for 7 subsystems (camera/radar perception, localization, planning, motion control, grasping, communications, safety), measuring recognition accuracy, drift, tracking error, and grasp success before closed-loop evaluation.
- Injected message delay, packet loss, sensor disconnection, and process restarts to measure fault-detection and recovery time; documented failure cases and drove fixes with algorithm, hardware, and systems teams.
- Hosted 3 technical demos for international partners that contributed to partnerships in Dubai and Singapore.

Shenzhen Zhi'ao Technology Co., Ltd. Shenzhen, China
Hardware Test Engineer Intern *Jul 2023 – Aug 2023*

- Tested Intel N97/N100/i3/i5 and Rockchip RK3568/RK3588 boards and industrial PCs on Windows and Ubuntu, covering boot, thermal throttling, DDR4/DDR5 and NVMe compatibility, and 24/48-hour sustained-load stability.
- Verified RS232/RS485 serial, Ethernet, Wi-Fi/Bluetooth, multi-display, and GPIO interfaces, writing C and Python test scripts for PLCs, industrial cameras, temperature sensors, and relays.
- Documented failures and partnered with hardware and software teams on fixes, contributing to a 15% reduction in hardware failure rates.

RESEARCH AND PROJECTS

Medical Robotics Research | NYU, Advisor: Prof. Rui Li Feb 2026 – Aug 2026

- Built an Arduino and Python (pyserial) data-acquisition workflow to stream, plot, and analyze two tactile sensors in real time.
- Characterized baseline noise and contact response, tuned detection thresholds from measured ADC ranges, and quantified cross-talk between channels during simultaneous operation.
- Evaluated waterproof encapsulation (PDMS, Ecoflex, Parylene C) and defined bare-vs-coated metrics for sensitivity, baseline stability, noise, and response time.
- Integrated a haptic device through ROS2 with 3D Slicer, RViz, and VMTK vascular models; owned driver integration and resolved USB-passthrough instability.

Embedded Gesture Recognition | Individual Project STM32L475, LSM6DSL, Embedded C

- Built a real-time gesture-recognition system from IMU data with UART output; verified signals and debugged communication faults with an oscilloscope.

Wanderland Navigation Benchmark | NYU AI4CE Lab, PI: Prof. Chen Feng Apr 2026 – Present

- Build HPC workflows for MetaCam data ingestion, point-cloud preprocessing, 3D Gaussian Splatting reconstruction, and quality control; lead student data collection.

Sb_2S_3 Thin-Film Solar Cells | Undergraduate Thesis 2024

- Fabricated thin-film devices and characterized them with XRD, SEM, J–V, and monochromator-based EQE spectral response; surface treatment raised efficiency from 4.43% to 5.43%.

Publication: First author, ICONE30 (2023). Verified the spatial-discretization accuracy of a Method of Characteristics reactor physics code using the Method of Manufactured Solutions.

SKILLS

Sensors & Test: data acquisition, ADC, noise and cross-talk analysis, oscilloscope, monochromator/EQE, fault injection
Embedded: STM32 (ADC, DMA, timers, UART, SPI, I2C), Arduino, embedded C, Keil, RS232/RS485, Ethernet
Programming: Python (pandas, NumPy, matplotlib), SQL, C, C++, MATLAB, Verilog, Linux, Git, ROS2